

di-
st-
fi-
ca-
tion
an-
pli-
fi-
ca-
tion
 $\times$
[2.28, 6.58]
??
 $\times$
 $\times$
??

First
 $\gamma =$
0.272
 $\bar{t} =$
3.18
 $\gamma =$
0.097
 $\times$
[2.28, 6.58]
 $\times$
 $\bar{r} =$
0.376
 $\bar{F} =$
58.8
 $p <$
0.001

Second
8.63/*VIX*
8.63/*VIX*
-13.7%
-33.8%
-21.2%
 $t =$
0.86
??
??
??
??
??
??
??
??
??
??

0050.TW
TWII
(TAIEX)
Nine
in-
di-
vid-
ual
Tai-
wan
stocks
and
one
ETF

??
U.S.
bench-
marks
Asia-
Pacific
mar-
kets
VIXTWN

$r_t =$
 $\ln(P_t/P_{t-1}) \times$
100
??
 $\gamma_{GJR}t(\gamma)$
-0.31
-0.47
-0.52
-0.18
-0.21
-0.22
Notes:
 γ_{GJR}
 $\bar{y} =$
2000
 $t(\gamma)$
??

**GJR-
GARCH(1,1)**

$$\sigma_t^2 = \omega + (\alpha + \gamma \cdot 1_{r_{t-1} < 0}) r_{t-1}^2 + \beta \sigma_{t-1}^2$$

(1) $\gamma >$
0
GARCH(1,1)
 $\gamma =$
0
EWMA

(Ex-
po-
nen-
tially
Weighted

$$\frac{r_t^f}{t-1}$$

3
??

$r_{market} <$
 0
 $r_{market} >$
 0
 $\times$
 $[2.28, 6.58]$
 $\times$
 $N =$
 9
 $[2.28, 6.58]$
 $B_{\overline{1,000}}$

4
cross-
market
within-
market

-33.8
-21.2
-22.2
-22.2
-13.7
t

Notes:
45496c00
q
3,968
n,511
t
-41.3%
??
t=
5

??
8.63/VIX
??
8.63/VIX
8.63/VIX
8.63/VIX

??⁵
di-
tional
change

+32%
+69%
~